CM3191 Inorganic Experiment 2: Synthesis and Characterisation of Ferrocene

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Short Lab Report • Thursday laboratory, fume hood B7 • 10 and 17 September 2026
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1. Aims

To prepare ferrocene, $Fe(\eta^5\text{-}C_5H_5)_2$, from cyclopentadiene, KOH and $FeCl_2 \cdot 4\,H_2O$ under nitrogen, purify it by sublimation, and establish its identity, purity and electronic structure by stereomicroscopy, ^{1}H NMR, ATR-IR and UV–Vis spectroscopy, ESI-MS and magnetic susceptibility, each interpreted against the 18-electron, low-spin d^6 sandwich description.[1]

2. Results and Discussion

2.1 Synthesis, yield and purification

Cyclopentadiene (1.0 mL, 0.80 g, 12.10 mmol) was deprotonated by KOH (4.0170 g, 71.6 mmol) in anhydrous DME (12 mL) under nitrogen and treated dropwise with $FeCl_2 \cdot 4\,H_2O$ (1.3274 g, 6.677 mmol) in DMSO (15 mL plus a rinse, as in the manual); quenching on ice with concentrated HCl, filtration, water washes and vacuum drying gave the crude product.[1] The overall reaction,

$$2\,C_5H_6 + 8\,KOH + FeCl_2 \cdot 4\,H_2O \longrightarrow Fe(C_5H_5)_2 + 2\,KCl + 6\,KOH \cdot H_2O, \tag{1}$$

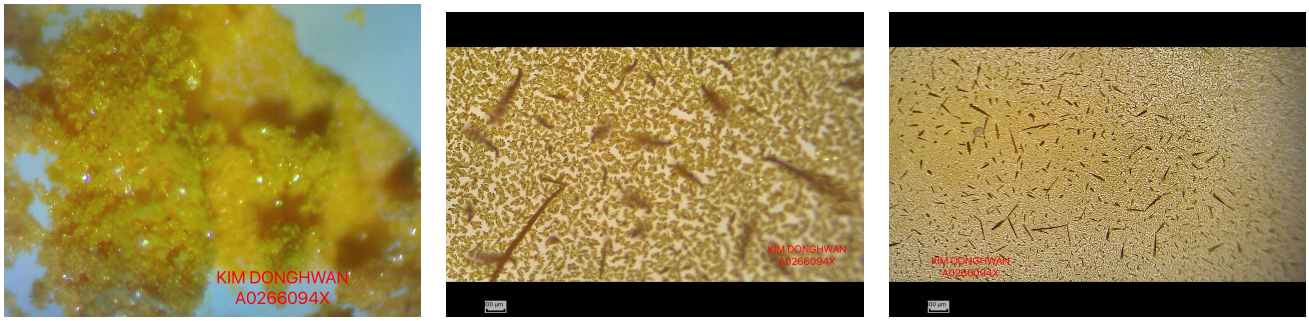
is a deprotonation (pK_a 18 for CpH, driven by the excess of KOH) followed by metathesis of two $C_5H_5^-$ with Fe^{2+}. Cyclopentadiene is limiting (iron in 10% excess), so the theoretical yield is 6.051 mmol × 186.04 g mol^{-1} = 1.126 g (Appendix A). The crude product weighed 1.6127 − 1.5428 = 0.0699 g, a crude yield of only 6.2%. The cyclopentadienide solution was only "pinkish brown" after 10 min rather than deep burgundy, which is consistent with incomplete deprotonation when the iron salt was added (Appendix D); the green colour on quenching is expected in any case, since iron was in excess.

Sublimation at 100–140 °C with an ice-cooled cover gave 0.0403 + 0.0123 = 0.0526 g of orange crystalline ferrocene from the original crude, a recovery of 75% and an isolated yield of 4.7% on the nominal cyclopentadiene charge. After the microscopy, NMR, ESI-MS and IR aliquots only 9 mg remained, less than the 20–25 mg needed for UV–Vis and the 0.10–0.12 g for the magnetic balance, so a second portion of crude (0.3021 g, weighed on tared paper; Appendix E) was issued and sublimed: 0.1104 + 0.0563 + 0.0138 = 0.1805 g (59.7% recovery). The UV–Vis and magnetic samples came from this second sublimate (Table 1); the yield refers to the original synthesis only. Because the crude yield alone caps the isolated yield at 6.2%, the low overall yield was dominated by incomplete product formation or losses before sublimation; purification losses (75% and 60% recoveries) made a smaller contribution. If the sublimate is essentially pure, the 75% recovery also implies that at least three-quarters of the crude was ferrocene, unless water condensed on the cycle-1 cover was weighed with it.

Table 1: Masses, yields and sample provenance (analytical balance, ±0.0001 g; Appendix D).

Quantity	Mass / g	Yield or recovery	Used for
Theoretical ferrocene (CpH limiting)	1.1258	—	
Crude ferrocene (1.6127 − 1.5428)	0.0699	6.2%	sublimation, microscopy
Sublimate from the original crude (cycles 1, 2)	0.0526	75.3% rec.; 4.7% overall	microscopy, NMR, ESI-MS, IR
Second crude portion	0.3021	—	
Sublimate from the second portion (cycles 3, 4, dish)	0.1805	59.7% rec.	UV–Vis (0.0232 g), MSB (0.0867 g)

Under the stereomicroscope the crude is a compact mass of yellow–orange granules without defined faces, as expected for a rapidly precipitated powder, whereas both sublimates consist of discrete, flat, angular orange plates and needles with sharp edges, (Figure 1b,c), a change of habit consistent with recrystallisation by sublimation. The two sublimates look alike, but only the first was characterised by NMR, IR and ESI-MS; the UV–Vis and magnetic results assume the same composition for the second.



(a) crude, Session I

(b) sublimed, original batch (100 μm bar)

(c) sublimed, second portion (100 μm bar)

Figure 1: Stereomicrographs at the manual's magnification of (a) the crude ferrocene, (b) the sublimate of the original crude (used for NMR, ESI-MS and IR) and (c) the sublimate of the second crude portion (used for UV–Vis and the magnetic balance); original files with embedded scale bars. A higher-magnification field of (c) is shown in Figure 4.

2.2 ^{1}H NMR spectroscopy

The spectrum of the sublimate (ca. 1 mg in 0.6 mL $CDCl_3$; 400.23 MHz, 297.9 K) is shown in full with every signal integrated in Figure 2, with the data-set name `CM3191-Thurs-400-Sep17-2026-09` on the plot; the axis is referenced to residual $CHCl_3$ at 7.26 ppm.[2] Signals, integrals relative to the ring singlet set to 10: δ 7.26 (s, 1.71, $CHCl_3$); **4.17 (s, 10.00, Fe(C_5H_5)$_2$)**; 2.17 (s, 0.21, acetone); 1.55 (s, 3.73, H_2O); 1.25 (br s, 0.13, aliphatic); literature 4.15 (s).[3]

A single sharp line (FWHM 2.0 Hz) for all ten ring protons is what the sandwich structure requires: the two η^5-rings are equivalent and rotate freely, so all C–H are identical on the NMR time scale. The weak lines at 4.389 and 3.953 ppm are the ^{13}C satellites of the singlet, not impurities: they are symmetric about it, separated by $^{1}J_{CH}$ = 175 Hz, and their combined integral, 1.2% of the parent, agrees with the 1.1% expected from the natural abundance of ^{13}C at each directly bonded carbon, $p/(1-p)$ with p = 0.0107 (Figure 5). No signal attributable to dienes (5–7 ppm), DME (3.40, 3.55) or DMSO (2.62) is observed above the noise; without an internal standard no quantitative detection limit is assigned. Besides the solvent signals, a small unassigned aliphatic signal (1.25 ppm) is present. The water (1.9 mol per mol ferrocene, about 0.2 mg in the tube) and acetone most likely entered with the solvent and the acetone-rinsed glassware; their origin cannot be assigned to the solid without a weighed sample and a solvent blank.

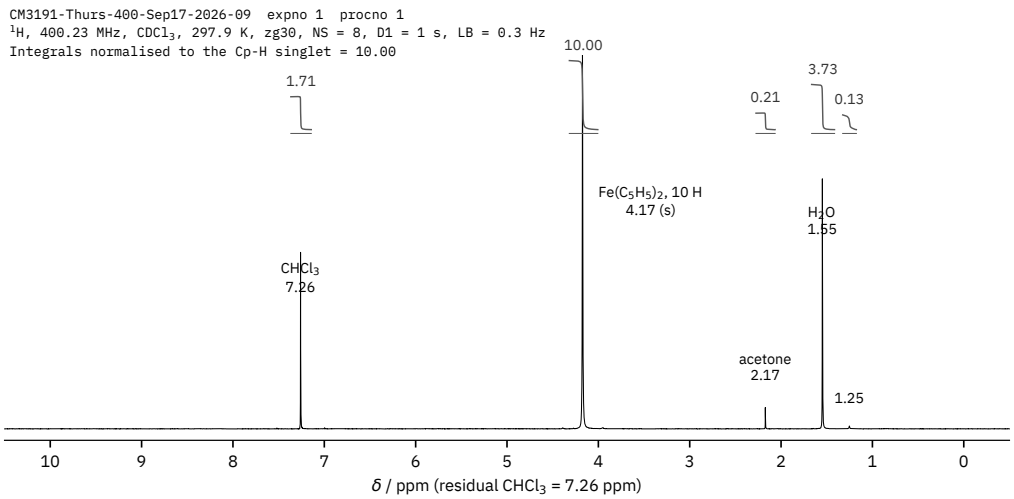


Figure 2: Full 0–10 ppm ^{1}H NMR spectrum of the sublimed ferrocene with all signals integrated (Cp-H = 10.00) and the data-set name shown; 400.23 MHz, $CDCl_3$, 297.9 K, zg30, 8 scans, D1 = 1 s, LB = 0.3 Hz; referenced to $CHCl_3$ = 7.26 ppm. Expansions in Figure 5.

2.3 ATR-IR spectroscopy

The solid-state spectrum (500–4000 cm^{-1}, Figure 6) shows five characteristic ferrocene bands in the recorded range, each within 10 cm^{-1} of the Lippincott–Nelson assignments:[4] 3095 (lit. 3085, ring C–H stretch; no aliphatic C–H below 3000), 1407 (1411, antisymmetric C–C stretch), 1104 (1108, ring breath-

ing), 999 (1002, in-plane C–H bend) and 812 cm^{-1} (811, out-of-plane C–H bend, the strongest band). No broad O–H absorption appears above 3200 cm^{-1}. Weak bands (dips of a few % transmittance) at 1631–1775, 1255, 1190, 1047, 855 and 786 cm^{-1} are not assigned here: ferrocene has weak overtone and combination bands, but a minor impurity absorbing near 1600–1800 cm^{-1} cannot be excluded from this spectrum alone.

2.4 UV–Vis spectroscopy and ligand-field assignment

Ferrocene (0.0232 g) in methanol (25.00 mL), $c = 0.0232/(186.04 \times 0.02500) = 4.988 \times 10^{-3}$ M, was measured undiluted in a 1.00 cm cell, both maxima falling inside the 0.1–1.0 absorbance window ($D = 1$): 440.50 nm ($A = 0.416$) and 325.00 nm ($A = 0.239$) (Figure 7). From $A = \varepsilon c l$,

$$\varepsilon_{440} = \frac{0.416}{4.988 \times 10^{-3} \times 1.00} = 83\ \mathrm{M^{-1}\,cm^{-1}}, \qquad \varepsilon_{325} = \frac{0.239}{4.988 \times 10^{-3} \times 1.00} = 48\ \mathrm{M^{-1}\,cm^{-1}}. \tag{2}$$

Sohn, Hendrickson and Gray report 22 700 cm^{-1} (440.5 nm, ε 91) and 30 800 cm^{-1} (324.7 nm, ε 49) in EPA: the positions agree within 1 nm, the absorptivities to 9% and 2%.[5,6] Absorptivities below 100 mark both bands as Laporte-forbidden d–d transitions. In the ligand-field assignment of Sohn *et al.* (D_{5d} labels; ground state $^1A_{1g}$, $(e_{2g})^4(a_{1g})^2$) the 440 nm envelope contains the two spin-allowed transitions $^1A_{1g} \rightarrow a\,^1E_{1g}$ and $^1A_{1g} \rightarrow {}^1E_{2g}$, and the 325 nm band is $^1A_{1g} \rightarrow b\,^1E_{1g}$; all three move an electron from the filled, essentially metal e_{2g}/a_{1g} orbitals into the empty antibonding e^*_{1g} level.[5] The two transitions within the 440 nm envelope are not resolved here. The 9% shortfall at 440 nm exceeds an illustrative uncertainty budget (about 1%, Appendix A); a non-absorbing impurity alone would lower both absorptivities by the same factor (observed 0.92 and 0.98 of the literature), so the solvent (methanol here, EPA in the literature) and the baseline may contribute, but the present measurement cannot separate these causes.

2.5 ESI-MS

The positive-ion electrospray spectrum of the sublimate (sample SN9; Thermo LTQ XL ion trap, flow injection in methanol, capillary 43 V, no added electrolyte; m/z 70–1400; Figure 3) has one dominant ion at m/z 185.96 with satellites at 184.04 (6.5%), 186.96 (14.2%) and 188.04 (1.2%); nothing else exceeds 5%.[10] The dominant ion is assigned to the ferrocenium radical cation $[Fe(C_5H_5)_2]^{+\bullet}$ rather than to $[M + H]^+$ or to $[M + Na]^+$ (209, absent, although sodium is present in the system); easily oxidised analytes such as ferrocene are known to be oxidised electrochemically in the electrospray source.[9] The envelope calculated for $C_{10}H_{10}Fe^+$, m/z 184 (6.4%), 185 (0.7), 186 (100), 187 (13.2), 188 (1.1), agrees with the observed cluster to within about one percentage point (Appendix B):[7] the $M-2$ component matches ^{54}Fe, and the $M+1$ peak is accounted for mainly by ^{13}C (10.8%) and ^{57}Fe (2.3%) relative to the base peak. No dimer (373), Cp-loss fragment (121) or substituted-ferrocene ion is detected. The spectrum supports the formula $C_{10}H_{10}Fe$; ESI intensities are not quantitative for composition.

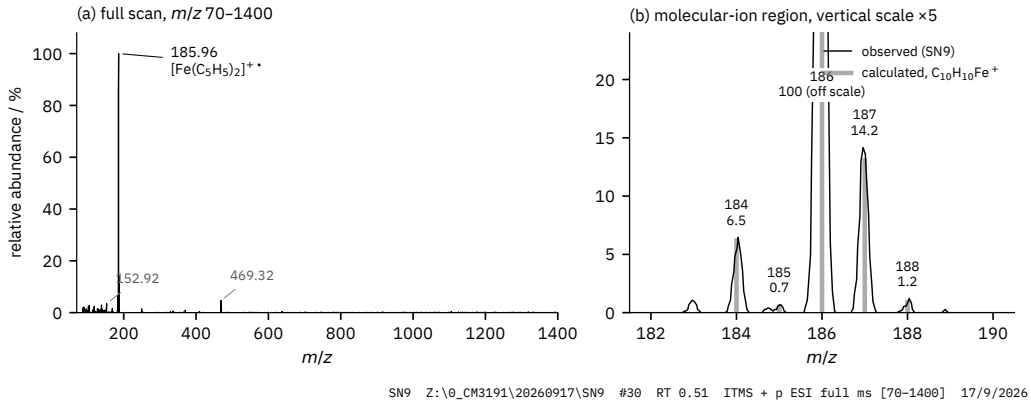


Figure 3: Positive-ion ESI mass spectrum of the sublimed ferrocene (SN9): (a) full scan; (b) molecular-ion region with the envelope calculated for $C_{10}H_{10}Fe^+$ (grey bars). Re-plotted from the instrument export; original printout in Appendix C.

2.6 Magnetic susceptibility

The ground sublimate was packed to 17 mm in the tube of a Sherwood MSB AUTO (unit 1) and a sample mass of 0.0867 g was entered; the tare record is inconsistent, so this net mass could not be independently verified (Appendix A). The displays read $\chi_v = -0.005\times10^{-4}$ and $\chi_g = 0.000\times10^{-4}$ $cm^3\ g^{-1}$; the instrument-

room temperature recorded by the laboratory staff was 23 °C (296 K).[10] The five photographs required by the procedure are Figure 8.

The literature value $\chi_M = -125 \times 10^{-6}$ cm^3 mol^{-1}[6] corresponds to $\chi_g = -6.7 \times 10^{-7}$ cm^3 g^{-1}, which the mass mode should display as -0.007×10^{-4}; it displayed 0.000. The two modes are also mutually inconsistent: since $\chi_g = \chi_v/\rho$ and a packed powder is less dense than the crystal (ca. 1.5 g cm^{-3}), the volume reading implies a mass-mode display of -0.003×10^{-4} or below. The initial zero read zero, but with no repeat zero, blank or calibration standard recorded, the measurement is inconclusive and no numerical susceptibility, moment or upper limit is assigned. Neither reading is positive, so the data give no indication of paramagnetism (one unpaired electron would display about $+0.068 \times 10^{-4}$ at 296 K), but they do not independently establish the diamagnetism expected for closed-shell, low-spin, 18-electron d^6 ferrocene. Applying the Pascal correction ($+125 \times 10^{-6}$) to a displayed zero would only manufacture a spurious moment.[8] The packed mass was also slightly below the manual's 0.10–0.12 g.

2.7 Sources of error

Possible yield losses include incomplete deprotonation, re-dimerisation and evaporation of cyclopentadiene, air oxidation of Fe^{2+} during the slow addition, under-delivery of the 1.0 mL charge, product left in flask, funnel and filter, and losses on lifting the sublimation cover; the records do not apportion the loss among them. Water or KCl in the crude would inflate the crude yield and deflate the apparent recovery. For UV–Vis, the uncertainty budget in Appendix A uses typical tolerances rather than measured repeatabilities, and the literature values were obtained in a different solvent. The magnetic measurement lacked a blank and a calibration standard, and its tare record is inconsistent.

3. Conclusion

Ferrocene was synthesised from cyclopentadiene and $FeCl_2 \cdot 4\,H_2O$ via potassium cyclopentadienide and purified by sublimation in 4.7% isolated yield (6.2% crude; sublimation recoveries 75% and 60%), the low yield being dominated by losses before sublimation. For the original sublimate, the ^{1}H NMR spectrum (one sharp Cp-H resonance at δ 4.17 with its ^{13}C satellites), five characteristic ATR-IR bands and the ferrocenium ion at m/z 186 with its iron and carbon isotope pattern identify ferrocene as the predominant component; minor solvent signals, an unassigned aliphatic NMR signal and weak unassigned IR bands were also observed. The second sublimate showed ligand-field bands at 440.5 nm ($\varepsilon = 83$) and 325.0 nm ($\varepsilon = 48$), within 1 nm of the literature. The magnetic measurement was inconclusive: neither reading was positive, but the readings could not be reconciled with each other or with the literature diamagnetism. The data identify the product as ferrocene but do not establish its quantitative bulk purity; its closed-shell, low-spin d^6 structure rests on the spectroscopic data and the literature rather than on the susceptibility measurement.

References

[1] Chan, E. *CM3191 Chemical Synthesis Experiments: Lab Manuals for Inorganic Experiments*, version 0626; National University of Singapore, AY2026/27; Experiment 2, pp 12–22; Appendix (MSB AUTO), p 23.

[2] Fulmer, G. R.; Miller, A. J. M.; Sherden, N. H.; Gottlieb, H. E.; Nudelman, A.; Stoltz, B. M.; Bercaw, J. E.; Goldberg, K. I. NMR Chemical Shifts of Trace Impurities: Common Laboratory Solvents, Organics, and Gases in Deuterated Solvents Relevant to the Organometallic Chemist. *Organometallics* **2010**, *29*, 2176–2179.

[3] Powers, T. M. Structure of Ferrocene. *JoVE Science Education: Inorganic Chemistry*; JoVE: Cambridge, MA, 2017. `https://app.jove.com/v/10347` (accessed 8 October 2026).

[4] Lippincott, E. R.; Nelson, R. D. The Vibrational Spectra and Structure of Ferrocene and Ruthenocene. *Spectrochim. Acta* **1958**, *10*, 307–329.

[5] Sohn, Y. S.; Hendrickson, D. N.; Gray, H. B. Electronic Structure of Metallocenes. *J. Am. Chem. Soc.* **1971**, *93*, 3603–3612.

[6] Sohn, Y. S. *Electronic Structures of Metallocene Complexes*. Ph.D. Dissertation, California Institute of Technology, 1971; Part I, Table II; Part II, Table I (χ_M after Engelmann, *Z. Naturforsch.* **1953**, *8b*, 775). DOI: 10.7907/pg0q-y417.

[7] Coursey, J. S.; Schwab, D. J.; Tsai, J. J.; Dragoset, R. A. *Atomic Weights and Isotopic Compositions*, version 4.1; National Institute of Standards and Technology: Gaithersburg, MD. `https://physics.nist.gov/Comp` (accessed 7 October 2026).

[8] Bain, G. A.; Berry, J. F. Diamagnetic Corrections and Pascal's Constants. *J. Chem. Educ.* **2008**, *85*, 532–536.

[9] Van Berkel, G. J.; Zhou, F. Characterization of an Electrospray Ion Source as a Controlled-Current Electrolytic Cell. *Anal. Chem.* **1995**, *67*, 2916–2923. DOI: 10.1021/ac00113a028.

[10] Chan, E. CM3191 Inorganic Canvas announcements, 24 September 2026 (ESI-MS instrumental settings) and 26 September 2026 (temperature in the magnetic measurement). National University of Singapore.

Figures

Figure 4: Higher-magnification stereomicrograph of the sublimate of the second crude portion (60 μm bar; original file with embedded label). Supplementary to Figure 1c, which is at the magnification prescribed by the manual.

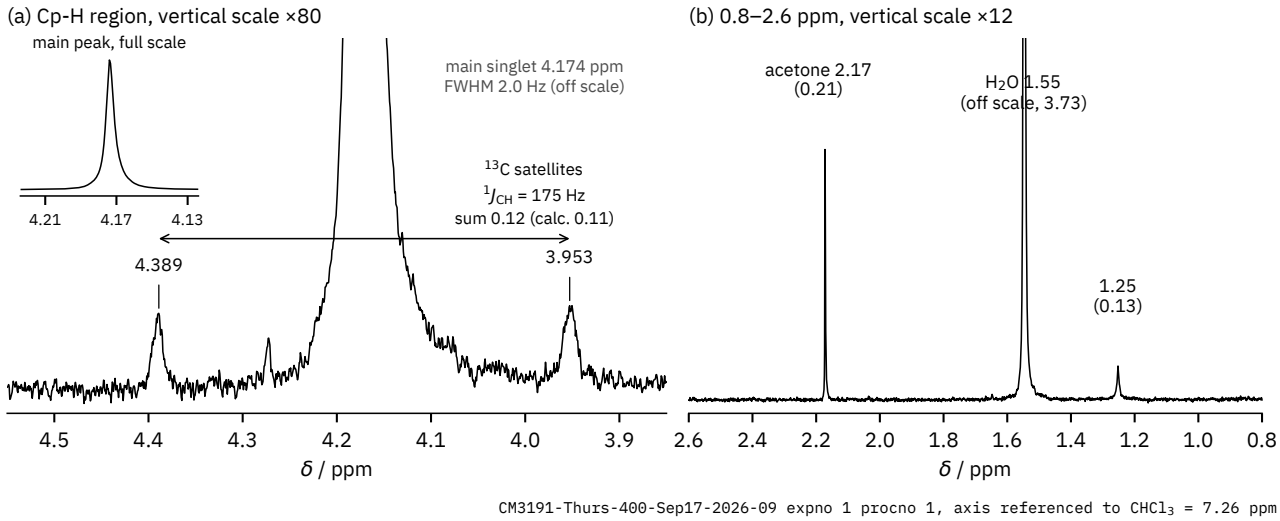


Figure 5: Expansions of Figure 2. (a) Cp-H region at 80-fold vertical gain showing the ^{13}C satellites ($^{1}J_{CH}$ = 175 Hz) and, inset, the parent singlet at full scale; (b) 0.8–2.6 ppm at 12-fold gain (integrals in brackets, Cp-H = 10.00).

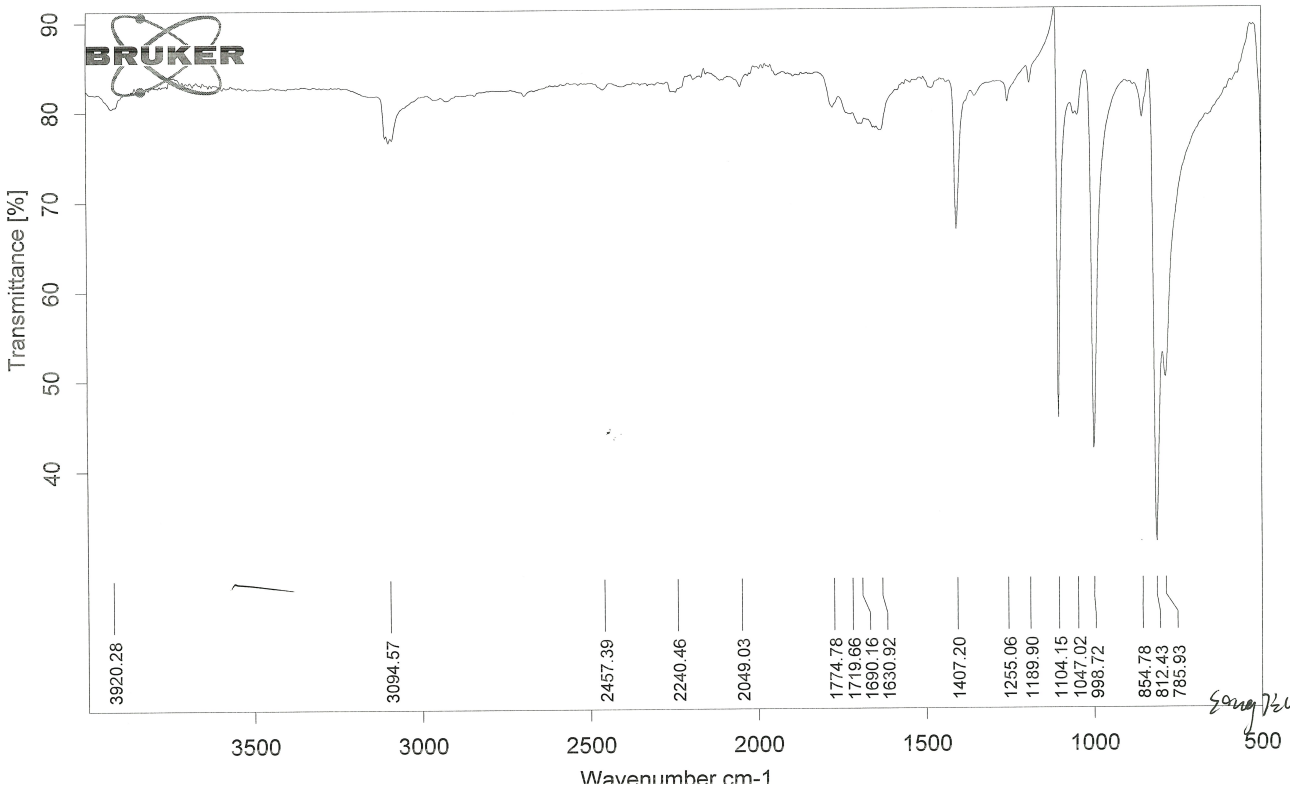


Figure 6: ATR-IR spectrum of the sublimed ferrocene, 500–4000 cm^{-1}, 17 September 2026 (detail of the signed printout; complete page in Appendix C).

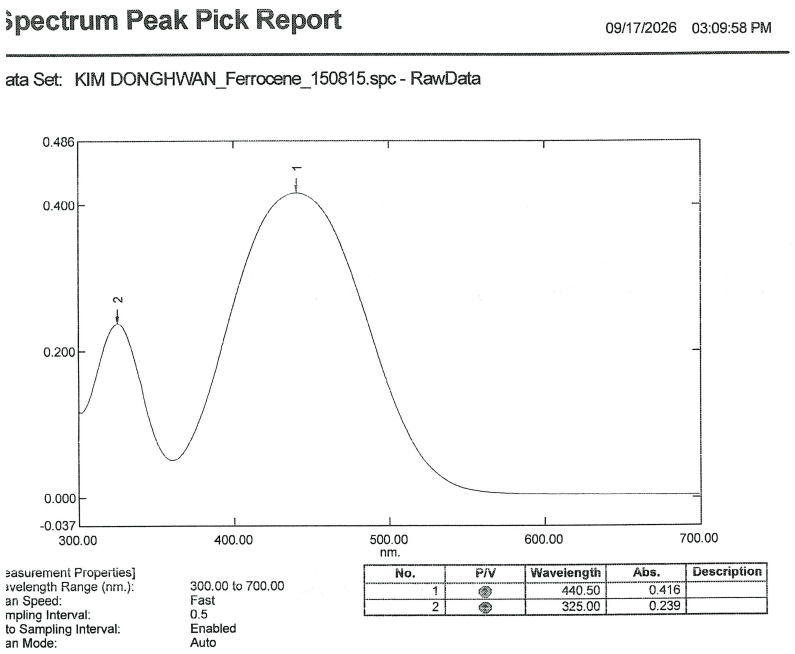
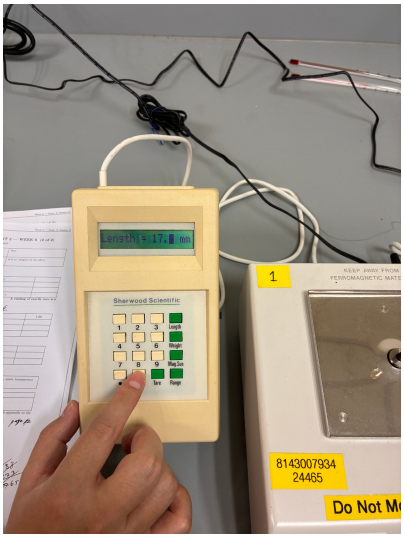
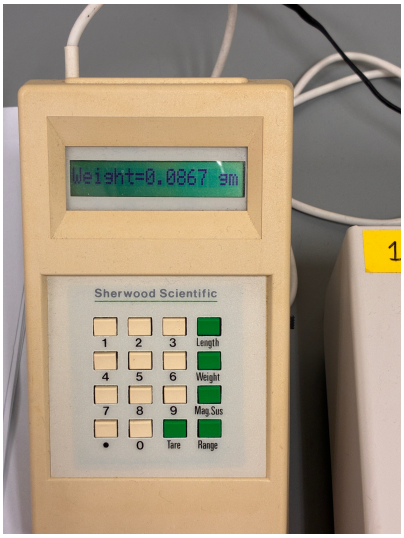


Figure 7: UV–Vis spectrum (300–700 nm, methanol, 1.00 cm) of the sublimed ferrocene, 4.988×10^{-3} M, with the instrument peak-pick table (440.50 nm, $A = 0.416$; 325.00 nm, $A = 0.239$); the signed printout with the handwritten ε calculation is in Appendix C.

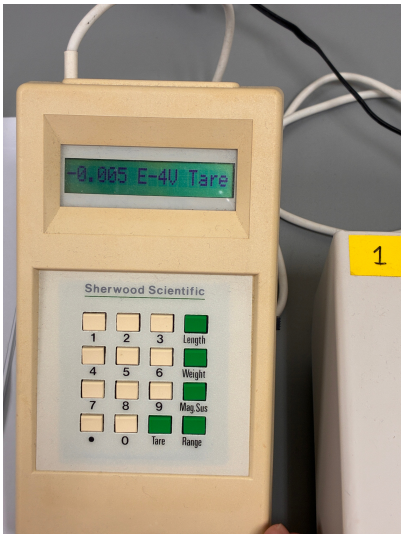
(1) zero reading before the tube was placed (`IMG_8576`; read as zero, but the photograph caught the display mid-refresh)



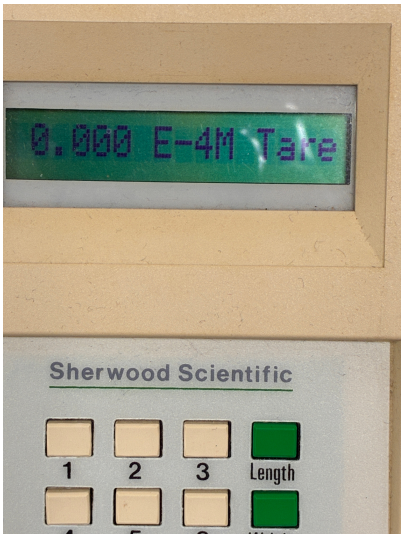
(2) length entry before the last digit, `17._ mm` (`IMG_8574`; detector labels unit 1 / 8143007934)



(3) weight entry, `Weight=0.0867 gm` (`IMG_8575`)



(4) volume susceptibility, `-0.005 E-4V` (`IMG_8577`)



(5) mass susceptibility, `0.000 E-4M` (`IMG_8579`)

Figure 8: The five photographs required by the manual for the Sherwood MSB AUTO measurement (unit 1), taken during the measurement under the demonstrator's supervision, in the order of the procedure: zero reading, length entry, weight entry, χ_v and χ_g results. Original, uncropped files. The weighing and packed-length photographs and enlarged display crops are in Appendix E.

Appendix A. Calculations

Limiting reagent and theoretical yield.

$$n(\mathrm{C_5H_6}) = \frac{1.0\ \mathrm{mL} \times 0.80\ \mathrm{g\,mL^{-1}}}{66.10\ \mathrm{g\,mol^{-1}}} = 12.10\ \mathrm{mmol}, \qquad n(\mathrm{FeCl_2 \cdot 4\,H_2O}) = \frac{1.3274}{198.81} = 6.677\ \mathrm{mmol}, \tag{3}$$

$$n(\mathrm{KOH}) = \frac{4.0170}{56.11} = 71.6\ \mathrm{mmol\ (8\ equiv = 48.4\ mmol)}, \qquad n(\mathrm{Fc})_{\mathrm{theor}} = \tfrac{1}{2}\, n(\mathrm{C_5H_6}) = 6.051\ \mathrm{mmol}, \tag{4}$$

$$m_{\mathrm{theor}} = 6.051 \times 10^{-3} \times 186.04 = 1.1258\ \mathrm{g} \quad \text{(CpH limiting; Fe in 10\% excess)}. \tag{5}$$

Yields and recoveries.

$$m_{\mathrm{crude}} = 1.6127 - 1.5428 = 0.0699\ \mathrm{g}, \qquad Y_{\mathrm{crude}} = 0.0699/1.1258 = 6.21\%, \tag{6}$$

$$m_{\mathrm{subl,orig}} = 0.0403 + 0.0123 = 0.0526\ \mathrm{g}, \qquad Y_{\mathrm{isolated}} = 0.0526/1.1258 = 4.67\%, \tag{7}$$

$$R_{\mathrm{subl,orig}} = 0.0526/0.0699 = 75.3\%, \qquad m_{\mathrm{residue}} = 1.1083 - 1.0989 = 0.0094\ \mathrm{g}, \tag{8}$$

$$m_{\mathrm{subl,2nd}} = 0.1104 + 0.0563 + 0.0138 = 0.1805\ \mathrm{g}, \qquad R_{\mathrm{subl,2nd}} = 0.1805/0.3021 = 59.7\%. \tag{9}$$

Molar absorptivity and uncertainty (undiluted stock, $D = 1$; tolerances are typical values, not measured repeatabilities).

$$c = \frac{0.0232\ \mathrm{g}}{186.04\ \mathrm{g\,mol^{-1}} \times 0.02500\ \mathrm{L}} = 4.988 \times 10^{-3}\ \mathrm{M}, \tag{10}$$

$$\varepsilon = \frac{A}{cl}\ (D = 1)\ : \qquad \varepsilon_{440} = 83.4, \qquad \varepsilon_{325} = 47.9\ \mathrm{M^{-1}\,cm^{-1}}, \tag{11}$$

$$\frac{u(\varepsilon)}{\varepsilon} = \sqrt{\left(\frac{u(A)}{A}\right)^2 + \left(\frac{u(m)}{m}\right)^2 + \left(\frac{u(V)}{V}\right)^2 + \left(\frac{u(l)}{l}\right)^2} = 0.7\%\ (440\ \mathrm{nm}),\ 1.0\%\ (325\ \mathrm{nm}), \tag{12}$$

with $u(A) = 0.002$, $u(m) = 0.0001$ g, $u(V) = 0.03$ mL, $u(l) = 0.002$ cm.

Magnetic susceptibility. The instrument was entered with a sample mass of 0.0867 g (IMG_8575) and the filled tube weighed 0.9090 g (IMG_8571). The datasheet holds two handwritten subtractions: $0.8988 - 0.8223 = 0.0765$ (crossed out) and a final one written as $0.9090 - 0.8264 = 0.0867$, which is arithmetically inconsistent ($0.9090 - 0.8264 = 0.0826$). The earlier tare, 0.8223 g, reproduces the entered mass ($0.9090 - 0.8223 = 0.0867$) and the 0.8264 entry is probably a copying error, but the final net mass cannot be independently verified from the records; the two candidates differ by 4.7%. The values below are comparison values only; no experimental susceptibility or moment is derived.

$$\text{displayed: } \chi_v = -0.005 \times 10^{-4}, \qquad \chi_g = 0.000 \times 10^{-4}\ \mathrm{cm^3\,g^{-1}}, \tag{13}$$

$$\chi_g(\text{from } \chi_v) = \chi_v/\rho \le -5 \times 10^{-7}/1.5 = -3.3 \times 10^{-7}\ \mathrm{cm^3\,g^{-1}}\ (\text{display } \le -0.003 \times 10^{-4}), \tag{14}$$

$$\chi_{g,\mathrm{lit}} = \frac{-125 \times 10^{-6}}{186.04} = -6.7 \times 10^{-7}\ \mathrm{cm^3\,g^{-1}}\ (\text{display } -0.007 \times 10^{-4}), \tag{15}$$

$$\chi_g(S = \tfrac{1}{2}) = \frac{\mu^2}{8TM} = \frac{1.73^2}{8 \times 296.15 \times 186.04} = +6.8 \times 10^{-6}\ \mathrm{cm^3\,g^{-1}}\ (\text{display } +0.068 \times 10^{-4}), \quad T = 23\ ^\circ\mathrm{C}. \tag{16}$$

Appendix B. Calculated ESI isotope envelope for $\mathrm{C_{10}H_{10}Fe^+}$

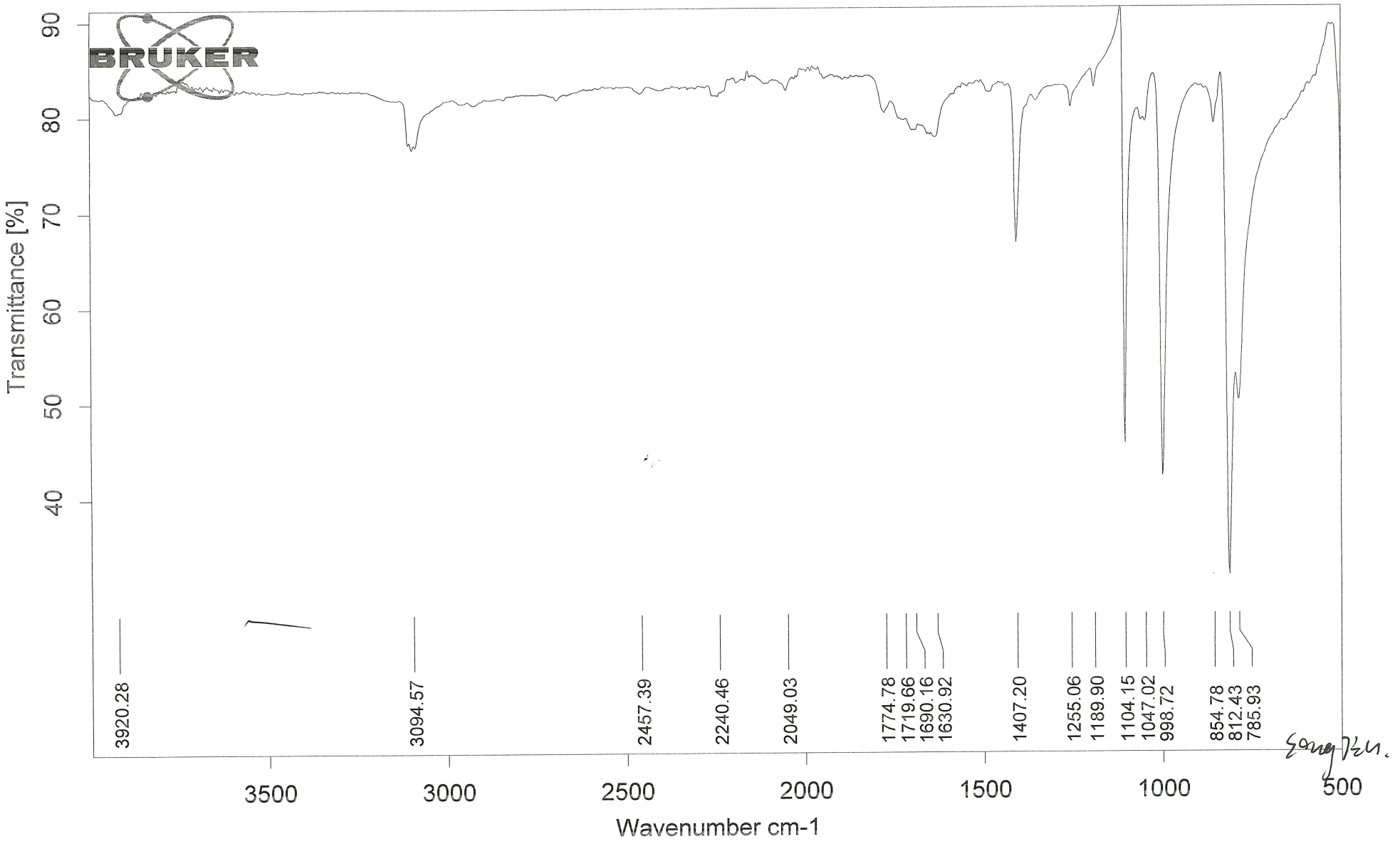
Natural-abundance distributions[7] convolved for ten C, ten H and one Fe; the ^{56}Fe group at nominal m/z = 186 is normalised to 100 (singly charged ion, no adducts). Observed values were read from the vector trace of the instrument printout (base peak = 100); m/z labels are the instrument's.

Nominal m/z	Calculated / %	Observed / %	Origin
184	6.37	6.5	^{54}Fe
185	0.70	0.7	^{54}Fe + ^{13}C
186	100.0	100	$^{12}\mathrm{C}_{10}\,^{1}\mathrm{H}_{10}\,^{56}$Fe
187	13.24	14.2	^{13}C (10.7%) or ^{57}Fe (2.1%)
188	1.10	1.2	$^{13}\mathrm{C}_2$, ^{58}Fe, or ^{13}C + ^{57}Fe
189	0.06	—	

First-order check: $I_{M+1}/I_M \approx 10\,x(^{13}\mathrm{C})/x(^{12}\mathrm{C}) + x(^{57}\mathrm{Fe})/x(^{56}\mathrm{Fe}) + 10\,x(^{2}\mathrm{H})/x(^{1}\mathrm{H}) \approx 0.108 + 0.023 + 0.001 = 0.132$.

Appendix C. Original IR, UV–Vis and ESI-MS printouts

The IR and UV–Vis pages are the signed originals; the concentration and ε calculation required by the manual is handwritten on the UV–Vis page. The ESI-MS pages are the instrument export of 17 September 2026 (file `SN9-KIM_DONGHWAN.docx`), rendered without alteration.

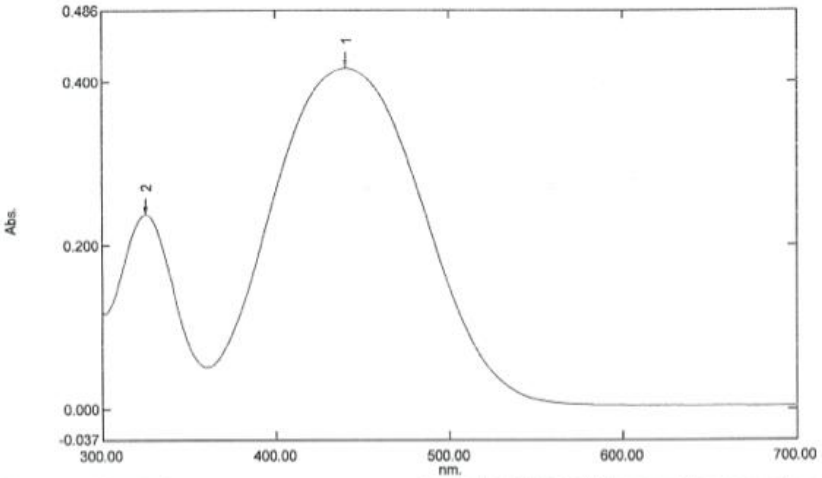


D:\CM3191\17Sep2026\Ferrocene_Kim_donghwan.0	Ferrocene_Kim_donghwan	Solid	17/9/2026

Spectrum Peak Pick Report

09/17/2026 03:09:58 PM

Data Set: KIM DONGHWAN_Ferrocene_150815.spc - RawData



No.	P/V	Wavelength	Abs.	Description
1	P	440.50	0.416	
2	V	325.00	0.239	

[Measurement Properties]
Wavelength Range (nm.): 300.00 to 700.00
Scan Speed: Fast
Sampling Interval: 0.5
Auto Sampling Interval: Enabled
Scan Mode: Auto

[Instrument Properties]
Instrument Type: UV-1800 Series
Measuring Mode: Absorbance
Slit Width: 1.0 nm
Light Source Change Wavelength: 340.0 nm
S/R Exchange: Normal

[Attachment Properties]
Attachment: None

[Operation]
Threshold: 0.0010000
Points: 4
InterPolate: Disabled
Average: Disabled

[Sample Preparation Properties]
Weight:
Volume:
Dilution:
Path Length:
Additional Information:

$M_{Ferrocene} = 186.04\ g/mol$.

$c = m/(MV)$

$= \dfrac{0.0232\ g}{\left[186.04\ \frac{g}{mol}\right] \cdot 0.02500\ L}$

$= 4.998 \times 10^{-3}\ mol/L$.

$A = \varepsilon c l \rightarrow A/cl = \varepsilon$.

$\lambda_1 = 440.50\ nm$.

$\varepsilon = \dfrac{0.416}{4.998 \times 10^{-3}\ mol\ L^{-1} \times 1.00\ cm}$

$= 83.4\ M^{-1} cm^{-1}$

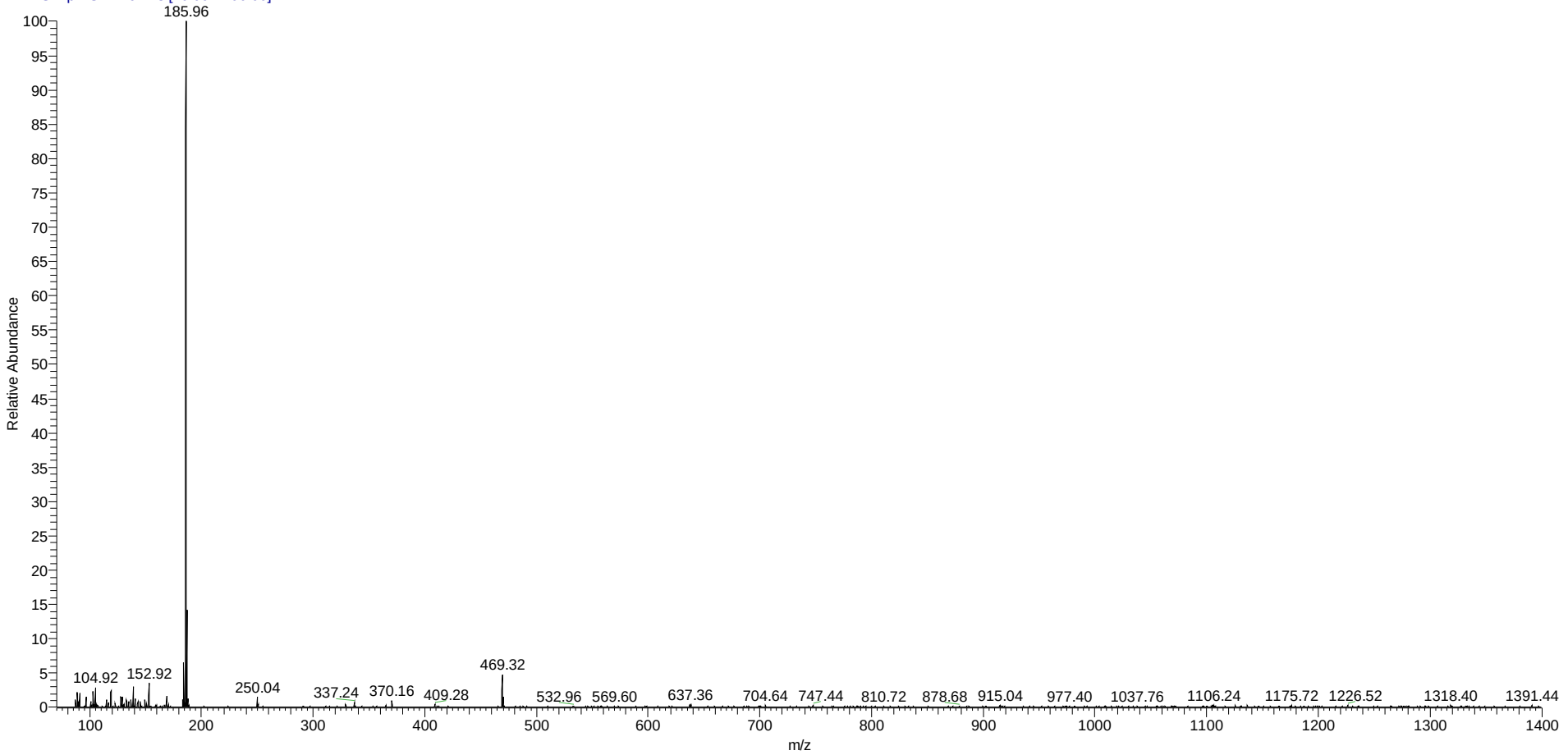
$\lambda_2 = 325.00\ nm$

$\varepsilon = \dfrac{0.239}{4.998 \times 10^{-3}\ mol\ L^{-1} \times 1.00\ cm}$

$= 47.9\ M^{-1} cm^{-1}$

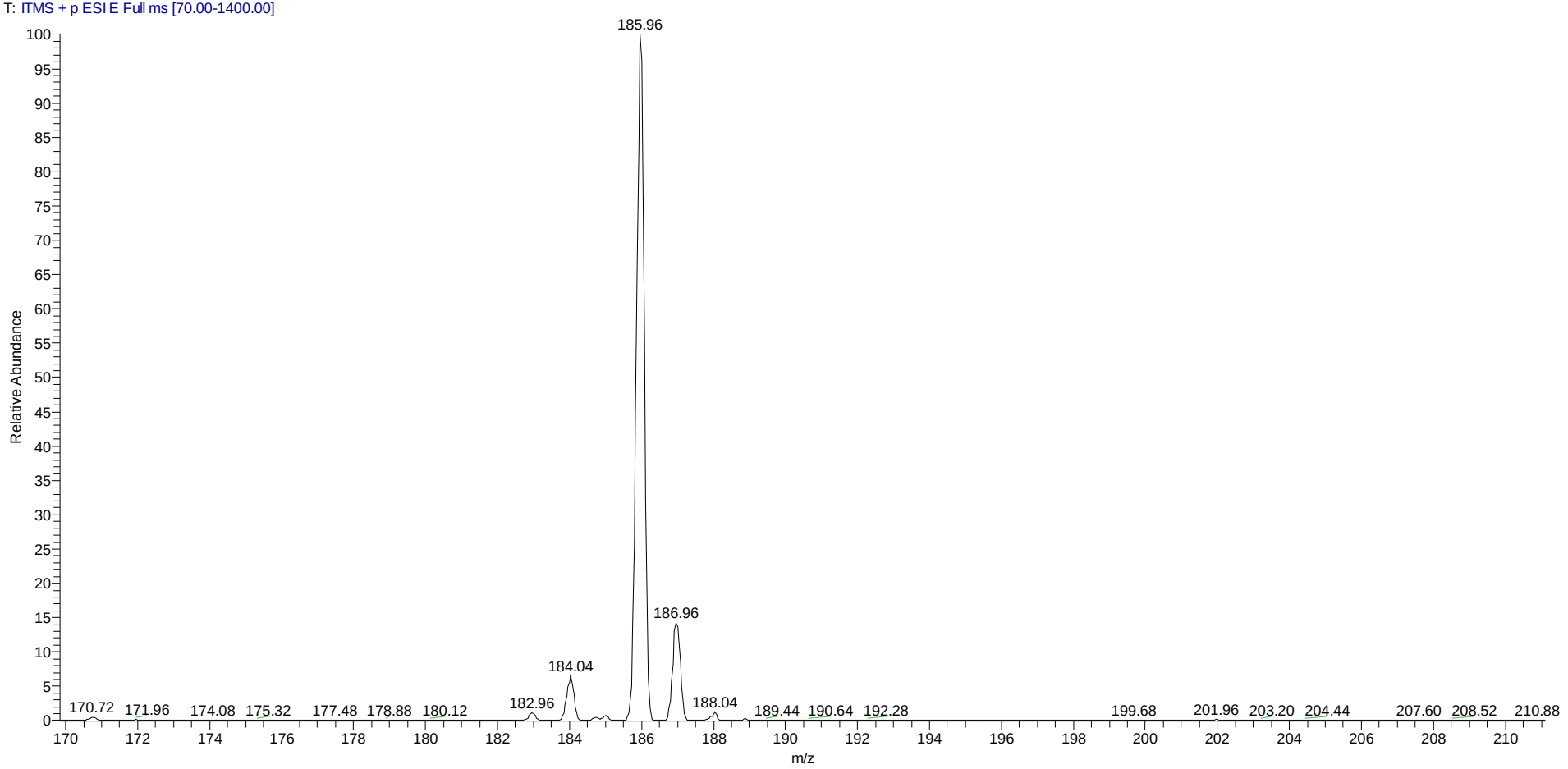
Z:\0_CM3191\20260917\SN9 17/9/2026 5:56:29 pm Enter Module Name
Enter Student Name

SN9 #30 RT: 0.51 AV: 1 NL: 1.41E5
T: ITMS + p ESI E Full ms [70.00-1400.00]



Z:\0_CM3191\20260917\SN9 17/9/2026 5:56:29 pm Enter Module Name
Enter Student Name

SN9 #30 RT: 0.51 AV: 1 NL: 1.41E5
T: ITMS + p ESI E Full ms [70.00-1400.00]



Appendix D. Signed datasheets

Reproduced without alteration (`TalkFile_Week5.pdf` p. 4; `TalkFile_Week6.pdf` pp. 2, 4, 5).

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 5

Session I: Synthesis and Microscopic Image of Crude Ferrocene

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

~30.03 g ice

Reagent	Nominal	Actual	Remarks
KOH pellets / anhydrous DME	4.0 g / 12 mL	4.0170 g	
Cyclopentadiene monomer	1.0 mL		
$FeCl_2 \cdot 4H_2O$ / DMSO (dissolve, rinse)	1.3 g / 7, 3.5 mL	1.3274 g	
Conc. HCl / ice	10 mL / 30 g		
DI water (rinse / wash 3×)	10 / 20 mL		

2. Conditions

	Nominal	Actual	Remarks
N_2 purge before CpH addition	3–5 min		
Stirring after CpH addition	10 min		
$FeCl_2$ addition rate / stirring after	20 drops min^{-1}, 30 min		
Stirring after acid quench	15–20 min		
Vacuum drying	~30 min		

3. Colour checkpoints — two of these tell you the reaction is working

Stage	Expected	Observed / time	✓
KOH in DME, before CpH	white undissolved solid, colourless liquid	(pale)	
10 min after CpH	**deep burgundy / red-brown.** A pale pink means deprotonation is incomplete	pinkish brown?	
$FeCl_2 \cdot 4H_2O$ in DMSO	**pale green.** Yellow or orange means Fe^{2+} is already oxidising		
During the $FeCl_2$ addition	darkens to a dark orange-brown slurry		
On quenching into ice-acid	orange precipitate; some gas evolution	green? + bright orange + some bubbling	
Filtrate and washings	pale yellow-green, fading to colourless		
Crude ferrocene, dried	orange to orange-brown powder		

The two bold rows are the checks worth making *before* committing the next reagent. A burgundy solution says the cyclopentadienide formed; a pale pink says stir longer. A green iron solution says the Fe^{2+} is still intact; if it has gone yellow or orange, oxidation has already cost you yield and the low result is explained. Write what you see, not what is printed.

4. Crude product

Mass of empty vial	1.5428.	N_2 flow maintained throughout ☐✓	ok
Mass of vial + crude	1.6127	Gas evolution on quenching ☐	slight?
Mass of crude ferrocene	0.07???	Microscopic image taken ☒	a/r/orange

To be completed at the end of the session		
Student signature	GTA name	GTA signature Song Yan 10/9/2026

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

2026/09/10

WEEK 6 (Mon. / Thu.)

INORGANIC EXPERIMENT 2 — SESSION II

Purification and Characterization of Ferrocene

Venue	General & Synthetic Chemistry Lab, Block S5 Level-1	**Instrument / technique**	Stereomicroscope, MSB, IR, UV–Vis, ESI-MS, ^{1}H NMR
Time	10:00 a.m. – 4:00 p.m. (Mon. or Thu.)	**Source**	Manual v0724 + lecture notes

(a) Purification by sublimation

1. **Weigh both larger petri dish covers empty** on the analytical balance and record each.
2. Spread the crude ferrocene on the smaller petri dish base; cover with a weighed cover.
3. Raise the hotplate *slowly*; **100 °C for cycle 1**, and never past **150 °C**. Place the 400 mL beaker of ice water on the cover.
4. Take the cover off the heat, let it cool, recover the sublimed ferrocene. Repeat until nothing further sublimes.
5. Weigh each loaded cover; transfer the product to a clean transparent bag or vial.

WHAT TO EXPECT FROM EACH CYCLE — SEE APPENDIX G

Cycle 1 mostly moves **water**: the crude is still damp from last week's filtration, so expect a wet, pale cover and little orange. Raise the temperature from cycle 2 or 3 and the orange flakes appear. Non-volatile impurities stay on the *base*, which darkens as the cycles proceed — that residue is the evidence that purification happened.
Let the cover cool before handling, invert it over the bag and not the bench, and re-weigh the cover empty before reusing it. Only two are issued.

(b) Characterization — six measurements

1. **Stereomicroscopy.** 14×, one image, matriculation number typed on it. Compare against the Week 5 crude image; this pair is the qualitative evidence for purification.
2. **Magnetic susceptibility.** Grind **0.09–0.12 g** in the agate mortar. Both χ_v and χ_g, five photographs. *Read Appendix H before computing a moment; Appendix I shows why the answer must be zero.*
3. **UV–Vis.** ~~35–40 mg~~ 20–25 mg, weighed to four decimals, in methanol, **25 mL volumetric flask**. Record 300–700 nm; dilute until $A = 0.1$–1.0. Determine ε for each peak; the calculation must be handwritten on the spectra.
4. **IR.** Solid spectrum 500–4000 cm^{-1}, ATR or KBr. Record which. *Appendix C.*
5. **^{1}H NMR.** **~1 mg** into a tube containing **0.6 mL $CDCl_3$**. One sharp singlet near δ 4.15 for all ten protons is the result — the ten C–H are equivalent because both rings rotate freely and all five carbons of each bind iron equally.
6. **ESI-MS.** ~~1–2 mg~~ 5mg into a sample tube. *Appendix B.*

THE MSB PROCEDURE — FIVE PHOTOGRAPHS, ALL ASSESSED

Switch on and wait for a zero reading (−0.000E−5Z) [**photo 1**]. Weigh an empty tube, place it on the detector, read, press TARE, confirm zero. Remove it, introduce the ground sample, measure the packed length (**15.0–25.0 mm**) and mass (**0.0010–0.9999 g**). Press LENGTH and enter three digits [**photo 2, before the last digit**]. Press WEIGHT and enter [**photo 3, before the last digit**]. Press MAG SUS and record χ_v [**photo 4**] and χ_g [**photo 5**]. The display advances the instant the last digit is entered.

Before you leave: ☐ Purified ferrocene labelled ☐ Microscope image ☐ Five MSB photos ☐ IR and UV–Vis spectra ☐ ESI-MS and NMR submitted ☐ Datasheet signed

original sample: microscopy ✓ → one more for extra sample → 1.4447 g initially (with vial)
^{1}H NMR sample submission ✓
ESI-MS sample submission
ATR-IR
UV-Vis.

1.1083
− 1.0989
(extra 20 mg)
→ new (extra sample

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 6 (1 of 2)

Session II: Sublimation Record

Full name (as on student ID)	Matriculation no.	Date
KZH DONGHUIGH.	A0266 094K	2026/09/17.
Lab session ☐ Mon ☑ Thu	Fumehood no. 137.	S/N no. assigned by lab officer

1. Starting material

petri s : 58.9177 g

Mass of crude ferrocene from Week 5		Hotplate setting, cycle 1	
Cover A, empty (g) Left	48.6311	Cover B, empty (g) right.	48.7280

2. Sublimation cycles

extra 45.7688

extra 48.6315

Cycle	Set / °C	Cover	Cover + Fc (g)	Fc (g)	Appearance of cover / base
1	~100°C	A / B	48.6714	0.0403	} 0.0526 g total
2	~140°.	A / B	45.7843	0.0123	
3		A / B	45.7792	0.1104	
4		A / B	48.6948.	0.0563	
			Total purified ferrocene (g)		58.9315 − 58.9177 = 0.0138

new (extra)

extra: 0.1805 g

3. Colour checkpoints

Stage	Expected	Observed / time	✓
Crude, before heating	orange to orange-brown powder		
Cover after cycle 1	**wet and pale** — mostly condensed water, little orange		
Cover after cycles 2–3	orange flakes or plates		
Petri dish **base**, later cycles	darkens; non-volatile residue left behind		
Purified product	bright orange crystalline flakes		
After grinding for the MSB	light orange free-flowing powder		
Methanol solution for UV	clear yellow-orange		

Expected is a checkpoint, not an answer. A pale cover after cycle 1 is the normal result, not a failure; record what you see and the time.

To be completed at the end of the session		
Student signature	GTA name	GTA signature Sang Jun 17/09/26

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

2026/09/17 page 1.

1-vial 0.0526 g

Submit extra sample vial two

extra : 0.3021 g.

after sublimation/purification :

empty vial : 1.0961 g

1.3982

1.1845

0.0137

0.0945

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 6 (2 of 2)

Session II: Characterization Record

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

4. Magnetic susceptibility — Sherwood MSB AUTO

Zero reading		Packed length L (mm)	17 mm
Empty tube mass (g)		Sample mass m (g)	0.0867
Volume susceptibility χ_v	−0.005	Mass susceptibility χ_g	0.000
Room temperature (K)		Sign of χ_g ☐ − ☑ 0 ☐ +	

Photographs: ☐ zero ☐ before last digit of length ☐ before last digit of weight ☐ χ_v ☐ χ_g **A reading of exactly zero is a resolution limit, not a moment — Appendix H.**

5. UV–Vis dilute until $A = 0.1 \sim 1.0$. → look for ε.

Mass weighed (g, four decimals)		Path length (cm)	1.00
Flask volume (mL) / solvent	25, MeOH	Concentration c (M)	
Dilution factor, if any		λ_1 (nm) / A_1	
Expected λ: ~440 and ~325 nm		λ_2 (nm) / A_2	

6. IR, NMR, ESI-MS and imaging

IR method ☑ ATR ☐ KBr		Mass for ^{1}H NMR (~1 mg)	
Microscope image of purified ☐		Mass for ESI-MS (1–2 mg)	
S/N no. for NMR		S/N no. for ESI-MS	

7. Yield and observations

Purified mass / crude mass, and the % recovered	Crude vs purified under the microscope — size, habit, transparency

To be completed at the end of the session		
Student signature	GTA name	GTA signature
		Cheng Thu 17/09/26

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

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photo - control unit and magnetic susceptibility serial #.

17 mm.

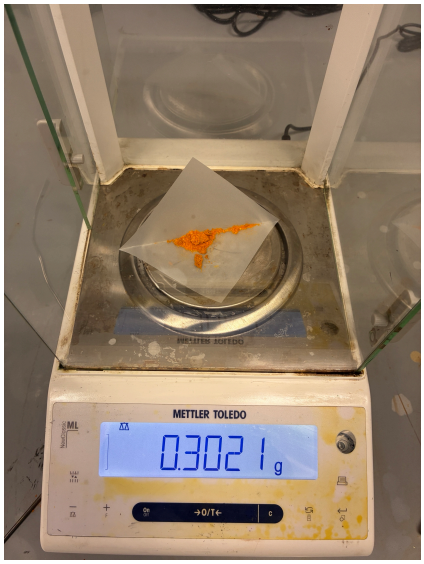
1. zero ②
2. ~~control~~ unit ⑦ length
3. weight. ⑧
4. control ⑨
5. balance.

total 5 photos.

0.8988
0.8223
0.0765 g.

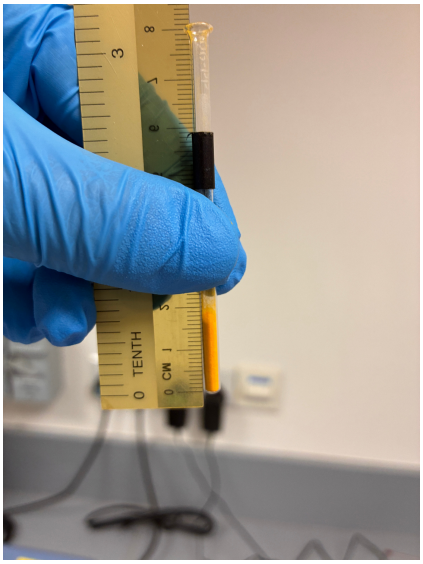
0.9090
− 0.8264
0.0826.

Appendix E. Weighing and magnetic-balance photographs



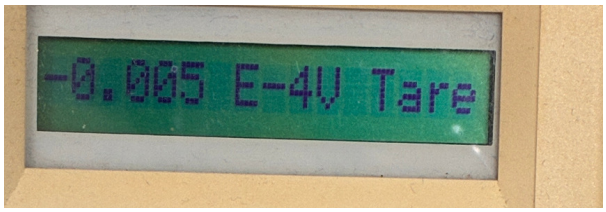
(a) second portion of crude ferrocene on tared weighing paper, 0.3021 g (IMG_8546)

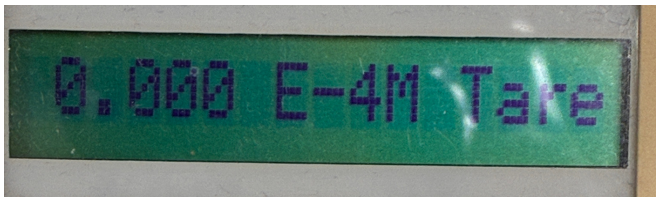
(b) filled MSB tube on the balance, 0.9090 g (IMG_8571)

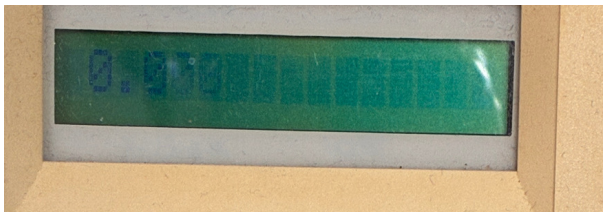


(c) packed sample beside the ruler, 17 mm (IMG_8572)

Figure 9: Weighing and packed-length photographs (original files).







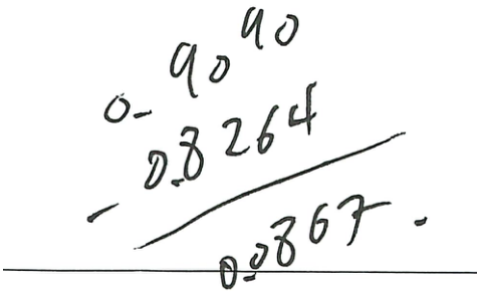


Figure 10: Enlarged display crops: volume-mode result, mass-mode result, the zero-reading frame (IMG_8576.JPG; the reading was zero, but the photograph caught the display while it was refreshing, so the digits are not legible) and the final handwritten mass subtraction on the datasheet, which reads 0.9090 − 0.8264 = 0.0867 (see Appendix A).